

On magnetic inhibition of photospheric macroturbulence generated in the iron-bump opacity zone of O-stars

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ABSTRACT

Massive, hot OB-stars show clear evidence of strong macroscopic broadening (in addition to rotation) in their photospheric spectral lines. This paper examines the occurrence of such ‘macroturbulence’ in slowly rotating O-stars with strong, organized surface magnetic fields. Focusing on the C IV 5811 Å line, we find evidence for significant macroturbulent broadening in all stars except NGC 1624–2, which also has (by far) the strongest magnetic field. Instead, the very sharp C IV lines in NGC 1624–2 are dominated by magnetic Zeeman broadening, from which we estimate a dipolar field ~ 20 kG. By contrast, magnetic broadening is negligible in the other stars (due to their weaker field strengths, on the order of 1 kG), and their C IV profiles are typically very broad and similar to corresponding lines observed in non-magnetic O-stars. Quantifying this by an isotropic, Gaussian macroturbulence, we derive $v_{\text{mac}} = 2.2 \pm_{2.2}^{0.9} \text{ km s}^{-1}$ for NGC 1624 and $v_{\text{mac}} \approx 20\text{--}65 \text{ km s}^{-1}$ for the rest of the magnetic sample.

We use these observational results to test the hypothesis that the field can stabilize the atmosphere and suppress the generation of macroturbulence down to stellar layers where the magnetic pressure P_B and the gas pressure P_g are comparable. Using a simple grey atmosphere to estimate the temperature T_0 at which $P_B = P_g$, we find that $T_0 > T_{\text{eff}}$ for all investigated magnetic stars, but that T_0 reaches the $\sim 160\,000$ K layers associated with the iron opacity bump in hot stars only for NGC 1624–2. This is consistent with the view that the responsible physical mechanism for photospheric O-star macroturbulence may be stellar gravity-mode oscillations excited by sub-surface convection zones, and it suggests that a sufficiently strong magnetic field can suppress such iron-bump generated convection and associated pulsational excitation.

Key words: convection – line: profiles – stars: atmospheres – stars: early-type – stars: magnetic field.

1 INTRODUCTION

Modern high-resolution, high-signal-to-noise optical spectroscopy of hot stars shows clearly that rotation is not the only macroscopic line-broadening agent operating in their photospheres. The additional broadening is typically seen over the complete spectral line, has a Gaussian-like shape and very large widths, $\sim 50 \text{ km s}^{-1}$, well in excess of the photospheric speed of sound, $\sim 20 \text{ km s}^{-1}$. Indeed, such very large ‘macroturbulence’¹ seems to be a ubiquitous feature

of OB-star atmospheres (Howarth et al. 1997; Lefever, Puls & Aerts 2007; Simón-Díaz & Herrero 2007; Simón-Díaz et al. 2010; Markova et al. 2011; Najarro, Hanson & Puls 2011; Bouret et al. 2012).

However, since early-type stars lack the vigorous surface convection associated with hydrogen recombination – which is responsible for such additional, non-thermal broadening in late-type stellar atmospheres (Asplund et al. 2000) – the physical origin of macroturbulence in hot stars remains unclear. A long-standing suggestion has been stellar oscillations (Lucy 1976), and recently Aerts et al. (2009) showed that the collective line-broadening effect of numerous low-amplitude gravity-mode (g-mode) pulsations can indeed mimic the large inferred values of Gaussian-like macroturbulence. Stellar structure theory predicts that this pulsational broadening excited by *core convection* becomes significant only in evolved massive stars, and while first results for OB supergiants

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¹ In stellar spectroscopy, one typically distinguishes between such *macro* turbulence, with scales longer than the photon mean free path (mfp), and *micro* turbulence, with scales shorter than this mfp. For O-stars, inferred micro turbulent velocities are normally of the order of a few km s^{-1} , i.e. much smaller than the macroturbulence that is the focus of this paper.

Table 1. Literature stellar and magnetic parameters and median macroturbulent velocities derived in this paper, including errors derived from the 68 per cent confidence regions.

Star	Obs. date	Inst.	Spec. type	T_{eff} (kK)	$\log g$ (cgs)	B_{pole} (kG)	P_{rot} (d)	$v \sin i$ (km s ⁻¹)	v_{mac} (km s ⁻¹)
^a NGC 1624–2	2012-09-27	ESP	O6.5–O8 f?p	35	4.0	20	158	0	2.2 ± ^{0.9} _{2.2}
^b HD 191612	2008-08-19	ESP	O6 f?p–O8 f?p	35	3.5	2.5	538	0	62.0 ± ^{0.5} _{0.5}
^c HD 57682	2010-01-29	ESP	O9 V	34	4.0	1.7	64	0	19.2 ± ^{0.3} _{0.3}
^d CPD 28 2561	2012-02-10	ESP	O6.5 f?p	35	4.0	1.7	70	0	24.3 ± ^{1.0} _{0.9}
^e HD 37022	2006-01-08	ESP	O7 Vp	39	4.1	1.1	15	24	42.9 ± ^{0.5} _{0.6}
^f HD 148937	2010-07-23	ESP	O6 f?p	41	4.0	1.0	7	45	54.0 ± ^{0.9} _{0.9}
^g HD 108	2009-07/2009-08	Nar	O8 f?p	35	3.5	0.5	1.8 × 10 ⁴	0	64.4 ± ^{0.4} _{0.4}
^h HD 36861	2007-21-21	ESP	O8 III((f))	35	3.7	0	–	45	50.0 ± ^{0.3} _{0.3}

^aWade et al. (2012b); ^bWade et al. (2011); ^cGrunhut et al. (2012); ^dBarba (in preparation), Petit et al. (2013);

^eWade et al. (2006); ^fWade et al. (2012a); ^gMartins et al. (2010); ^hNajarro et al. (2011).

do indeed seem to support a pulsational origin of macroturbulence (Simón-Díaz et al. 2010), observations indicate the presence of significant macroturbulent velocities in hot stars of *all* evolutionary classes (Markova et al. 2011; Najarro et al. 2011; Markova, in preparation; Simón-Díaz, in preparation). Recently, however, Shiode et al. (2013) (see also Cantiello et al. 2009) suggested that *near-surface* convection generated by the iron-bump opacity zone can trigger excited g modes in main-sequence O-stars as well, and that such pulsations might be the origin of their observed turbulent broadening.

A key empirical issue regards the actual quantification of macroturbulence. In particular, additional broadening due to a basically asymmetric process like stellar oscillations will affect the first zero in the Fourier power spectra of spectral lines, which typically is used to infer the projected stellar rotation (e.g., Simón-Díaz & Herrero 2007), and so make it difficult to disentangle and quantify the relative amounts of macroturbulent and rotational broadening present in the atmosphere (Aerts et al. 2009). Here, we circumvent this issue by focusing on O-stars with detected organized surface magnetic fields, using high-quality spectra collected within the Magnetism in Massive Stars project (MiMeS; Wade, Grunhut & MiMeS Collaboration 2012c). In particular, all these magnetic O-stars have very long measured rotation periods, likely because they have been spun down through magnetic braking by their strong stellar winds (see Petit et al. 2013); their (in most cases) negligible rotational broadening makes them ideal targets for the present study.

From studies of the chemically peculiar Ap stars, it is well known that the presence of a sufficiently strong surface magnetic field can stabilize the atmosphere against large-scale motions and suppress convectively induced turbulent line broadening (Landstreet, Ryabchikova, Kochukhov, private communication, see also, e.g., Michaud 1970 and Landstreet 1996). Building on these concepts, we compare here the narrow lines observed in NGC 1624–2 (measured surface dipolar field ~ 20 kG; Wade et al. 2012b) with the broader lines observed in other magnetic O-stars (surface fields on order ~ 1 kG), in an attempt to place first observational constraints on the depth of the source region for the enigmatic macroturbulence in hot main-sequence stars.

The paper is organized as follows: Section 2 presents the observational sample and describes our method for deriving macroturbulent velocities. Section 3 gives the results of the analysis, and Section 4 presents a simple model for interpreting magnetic inhibition of macroturbulence. Finally, Section 5 discusses the results, gives our conclusions and outlines directions for future work.

2 OBSERVATIONS AND METHOD

We select magnetic O-type stars with well constrained rotational periods and strong magnetic fields from the compilation by Petit et al. (2013). The sample, listed in Table 1, consists of seven magnetic stars, including NGC 1624–2 which hosts a field an order of magnitude stronger than any other known magnetic O-star (Wade et al. 2012b). For comparison purposes, we also include and analyse the non-magnetic star HD 36861 ($=\lambda$ Ori A), which has stellar parameters similar to those of our magnetic sample.²

For all sample stars, we retrieve high-quality spectra from the extensive MiMeS data base. The observations were obtained with the high-resolution ($R \sim 65\,000$) spectropolarimeter ESPaDOnS at the Canada–France–Hawaii Telescope, or with its twin instrument Narval located at the T  lescope Bernard-Lyot. We focus on analysing the magnetic O-stars’ low states, defined according to when the cyclic H   emission originating from their ‘dynamical magnetospheres’ (Sundqvist et al. 2012) is at minimum. This should minimize any contamination of the photospheric lines by magnetospheric emission (for a review of massive-star magnetospheric properties, see Petit et al. 2013). However, at the end of Section 3, we also comment briefly on line-profile variability between this low state and the corresponding high state (defined by maximum H   emission).

To obtain macroturbulent velocities, we analyse the photospheric C IV 5811    line, accounting also for rotational and magnetic broadening when needed. Surface magnetic field properties of OB-stars are generally derived from polarimetric measurements, as the Zeeman splitting typically is too small to be separated from other broadening mechanisms (see Donati & Landstreet 2009). In the optical, the Zeeman broadening is only $\sim 1\text{--}2$ km s⁻¹ kG⁻¹, i.e. in the current sample only NGC 1624–2 has a surface field strong enough to display significant magnetic broadening in the line profiles (see Fig. 1). Nevertheless, for consistency we include a proper treatment of the Zeeman effect for all stars, using the literature dipolar field values listed in Table 1. We note that, due to the sometimes unknown observer inclination angle i , these inferred field strengths are uncertain by up to ~ 50 per cent; this uncertainty, however, does not affect the macroturbulent velocities derived in this paper. The Zeeman pattern of the C IV $3s^2S_{1/2} \rightarrow 3p^2P_{1/2}$ transition, calculated under LS coupling, has two strong σ components and two well-spaced, weaker π components. This pattern, along with a high

² Spectropolarimetric observations available for this star constrain the dipolar surface field to be below 80 G (David-Uraz, in preparation).

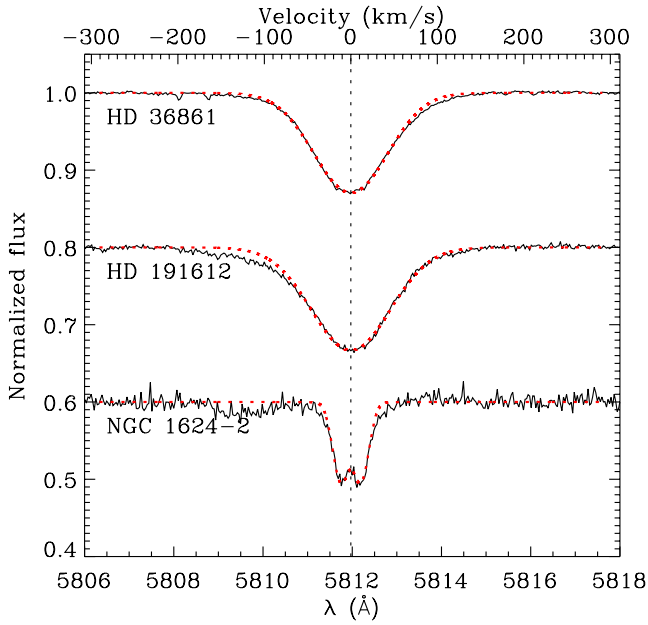


Figure 1. Observed (black solid) and fitted (red dashed) C IV line profiles for three stars in our sample, as labelled in the figure. The horizontal dashed line marks line centre and the continua in the two lower curves have been shifted downwards by 0.2 and 0.4 normalized flux units.

magnetic sensitivity (effective Landé factor $\bar{g} = 1.33$), makes this unblended line particularly suitable for the present analysis.

The rotation period of magnetic stars can be readily obtained from the observed variation of the longitudinal field (e.g., Borra & Landstreet 1980; Bychkov, Bychkova & Madej 2005) or photometric/spectral variations caused by their magnetospheres (e.g., Landstreet & Borra 1978; Howarth et al. 2007). Most of the magnetic O-stars have very long periods and so have negligible rotational broadening, except for HD 148937 and HD 37022 ($=\theta^1$ Ori C), for which we include broadening due to the modest rotation listed in Table 1.

We compute emergent intensities at each point on the stellar surface using the Unno–Rachkovsky solution for a Milne–Eddington atmosphere (Unno 1956; see also Landi Degl’Innocenti & Landolfi 2004, chapter 9). Synthetic flux profiles are then obtained by numerically integrating the emergent intensities over the projected stellar disc, assuming a dipolar surface field matching the simple geometry of the (presumed fossil) fields observed in these stars. Formally, the Zeeman splitting is sensitive to the surface field modulus; however, for equal disc-integrated values of the modulus, the actual position of the magnetic pole on the visible hemisphere does not change the structure of the line profile significantly. Therefore, we place the magnetic equator at the centre of the stellar disc for all stars, as the best representative viewing angle for their low states.

This simplified approach allows for a fast treatment of the Zeeman effect and derivation of macroturbulent velocities, and further avoids model dependences on e.g. stellar parameters and chemical abundances, which are not important for the analysis here. Within the Unno–Rachkovsky approximation, the source function S is linear in the continuum optical depth, $S(\tau_c) = S_0[1 + \beta\tau_c]$, and the polarized radiative transfer equations can be solved analytically. Specifically, we use $\beta = 1.5$, Voigt-shaped line profiles with damping constant $a = 0.02$ and a thermal speed $v_{th} = 10 \text{ km s}^{-1}$. We then fit the core of the line over a grid of profile strengths

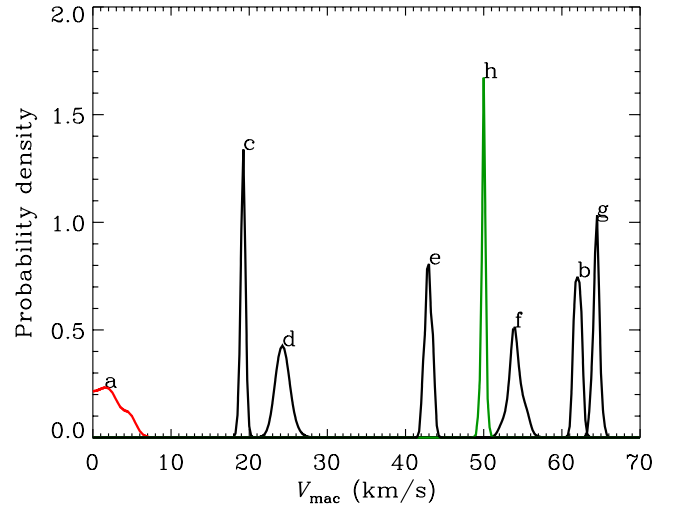


Figure 2. Marginalized probability density distributions of derived macroturbulent velocities, labelled according to Table 1. The red curve indicates NGC 1624–2 and the green curve indicates the non-magnetic comparison star HD 36861.

(set by the line-to-continuum opacity ratio) and isotropic.³ Gaussian macroturbulence of the form $e^{-v^2/v_{mac}^2}/(\sqrt{\pi}v_{mac})$. To estimate the admissible v_{mac} , we calculate the Bayesian probability density function marginalized over the profile strengths. The prior probability density is constant, and any deviations not explained by the model are treated as additional Gaussian noise, which results in the most conservative estimate of the parameters under the maximum entropy principle (e.g., Gregory 2005).

3 RESULTS

Using the method described above, Fig. 1 shows fitted C IV profiles for three stars in our sample, namely HD 191612, NGC 1624–2 and the non-magnetic comparison star HD 36861.

Since HD 191612’s rotation period is 538 d and its surface magnetic field is ‘only’ 2.5 kG (Table 1), both rotational and magnetic broadening are negligible for this star; thus, the total line broadening may be quite unambiguously associated purely with macroturbulence. Comparing this with HD 36861 reveals very similar Gaussian-like and broad C IV lines, indicating a common origin of the observed macroturbulence in magnetic and non-magnetic O-stars. By contrast, the observed line in NGC 1624–2 is qualitatively very different, much narrower and with magnetic Zeeman splitting directly visible (due to the very strong surface field). This indicates that the mechanism responsible for the large macroturbulent velocities in HD 36861 and HD 191612 is not effective in NGC 1624–2.

Fig. 2 further shows probability density distributions of macroturbulent velocities for the full stellar sample described in Section 2, and Table 1 lists the corresponding median values along with the 68 per cent confidence regions. For NGC 1624–2, we allow the (dipole) magnetic field strength to be a free parameter, in order to properly model the observed Zeeman splitting and obtain a more conservative estimate of the macroturbulent velocity. Fig. 3

³ In some studies (e.g., Simón-Díaz et al. 2010; Najarro et al. 2011) a radial-tangential formulation of macroturbulence is used rather than the isotropic one adopted here. In the former case, large-scale motions are assumed to occur only radially and/or tangentially to the stellar surface (see Gray 2005).

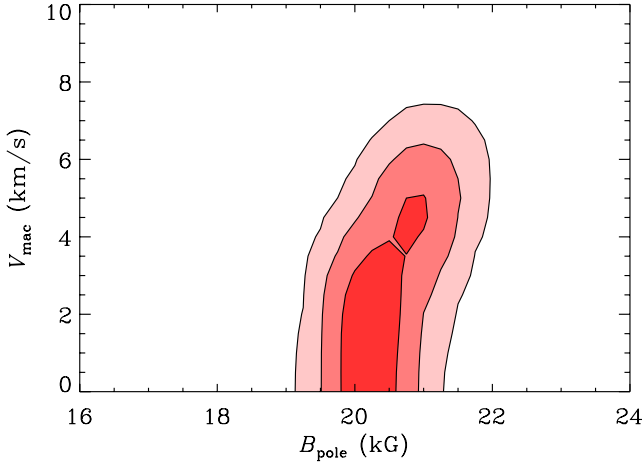


Figure 3. Joint probability-density contour map of derived macroturbulent velocity and magnetic dipolar field strength for NGC 1624–2. The contours encircle 68.3, 95.4 and 99.7 per cent of the probability.

illustrates the resulting joint-probability contour map of magnetic field strength and macroturbulence.

This simple analysis for NGC 1624–2 results in a very strong dipole field $B_{\text{pole}} \approx 20$ kG (consistent with Wade et al. 2012b) and a very low $v_{\text{mac}} = 2.2 \pm_{-2.2}^{+0.9}$ km s^{−1}. Such a low macroturbulent velocity is in stark contrast with the rest of the sample, which displays much higher values, $v_{\text{mac}} \approx 20\text{--}65$ km s^{−1} (see Fig. 2). Thus, among the known slowly rotating magnetic O-stars, macroturbulence is anomalously low in NGC 1624–2.

We have here focused on fitting profiles during the spectral ‘low state’ (see Section 2). Comparing the observed C IV lines in the low and high states of the magnetic stars indeed reveals variable and asymmetric profiles, but it is difficult to judge whether this variability is of photospheric origin or stems from refilling by magnetospheric emission during the high state. Thus, we defer a detailed investigation of such line-profile variability to a future paper, noting simply that the inferred macroturbulent velocities during the low and high states typically differ only by a few km s^{−1}, which does not affect our basic results or conclusions.

4 INTERPRETING MAGNETIC INHIBITION OF O-STAR MACROTURBULENCE

Let us now interpret the results of the previous section using a very simple model for magnetic inhibition of macroturbulence. Assuming the magnetic field stabilizes the atmosphere against large-scale motions approximately down to a stellar layer at which the magnetic pressure $P_B = B^2/(8\pi)$ equals the gas pressure P_g , we can analytically estimate the corresponding temperature T_0 by adopting a classical grey atmosphere with temperature structure at a given optical depth:

$$T(\tau) \approx T_{\text{eff}} \left(\frac{3}{4}\tau + \frac{1}{2} \right)^{1/4}. \quad (1)$$

For gravity g (in cm s^{−2}) and a nearly constant mass absorption coefficient κ (in cm² g^{−1}), the optical depth becomes

$$\tau = \int \kappa \rho dz \approx \kappa m_c = \kappa P_g / g, \quad (2)$$

where the last equality uses hydrostatic equilibrium to evaluate the column mass m_c . For a large-scale magnetic field with arbitrary, and

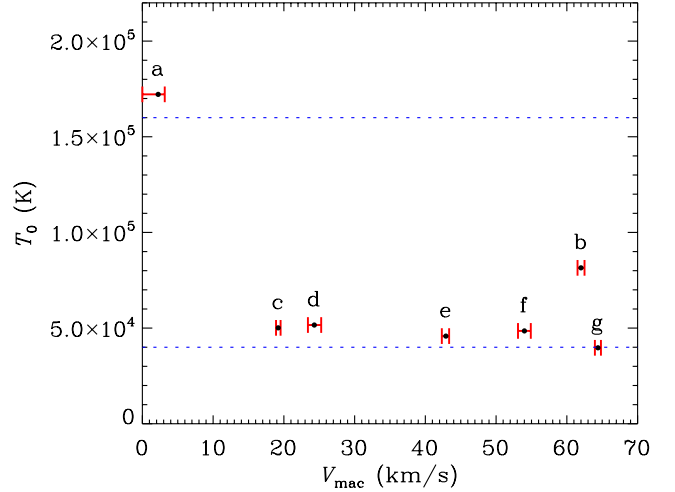


Figure 4. Atmospheric temperature T_0 (defined as the temperature at which gas and magnetic pressure equals according to the model described in the text) versus derived macroturbulent velocities with 1σ error bars. The stars are labelled according to Table 1. The dashed lines denote the approximate locations of the stellar effective temperatures (lower) and the iron opacity bump (upper).

possibly even variable, tilt to the vertical direction, the variation of density along a field line depends only on the vertical depth. With the further assumption that there is no horizontal variation in pressure or density between field lines, this standard hydrostatic condition between pressure and column mass still holds, i.e. $P_g = m_c g$.

Since the extent of the photosphere is very small compared to the stellar radius, we may further neglect the radial dependence of the magnetic field and use the inferred surface field strength B throughout the atmosphere.⁴ Setting then $P_g = P_B = B^2/(8\pi)$, one obtains the temperature T_0 in the layer at which the gas and magnetic pressure are equal,

$$T_0 = T_{\text{eff}} \left(\frac{3}{32\pi} \frac{B^2 \kappa}{g} + \frac{1}{2} \right)^{1/4} \approx 0.42 T_{\text{eff}} B^{1/2} (\kappa/g)^{1/4}, \quad (3)$$

where B has units of gauss. The second expression here neglects the $1/2$ within the parenthesis, and so implicitly assumes a field strength significantly stronger than the $B \approx 400 (10^{-4} g/\kappa)^{1/2}$ that yields $P_B = P_g$ at $T_0 = T_{\text{eff}}$.

Table 1 lists stellar and magnetic parameters used to estimate T_0 for the stars in our sample. We use the averaged surface field for B and, for simplicity, we further take $\kappa = 1 \approx 3\kappa_e$ for all stars, where $\kappa_e = 0.34$ is the electron scattering opacity for a hot star of standard solar composition. Inspections of Rosseland opacities in detailed FASTWIND non-local thermodynamic equilibrium model atmospheres (Puls et al. 2005) show that for atmospheric layers with $\tau_{\text{Ross}} \gtrsim 0.1$, such constant $\kappa \approx 3\kappa_e$ actually is a quite good opacity-estimate in Galactic O-stars that are not too evolved (see also, e.g., fig. 1 of Cantiello et al. 2009).

Fig. 4 displays T_0 versus the v_{mac} values derived in Section 3 for the magnetic O-stars. The figure shows that although the magnetic field likely affects the atmosphere of all investigated stars to some extent, this influence reaches down to much deeper layers

⁴ Also note here that since these large-scale, organized magnetic fields presumably are of fossil origin, they should not exhibit the increase in field strength with density that may be expected for fields in energy equipartition generated by stellar dynamos.

in the sharp-lined NGC 1624–2 than in any other star. A simple comparison of NGC 1624–2 ($v_{\text{mac}} = 2.2 \pm_{2.2}^{0.9} \text{ km s}^{-1}$) with HD 191612 ($v_{\text{mac}} = 62.0 \pm_{0.5}^{0.5} \text{ km s}^{-1}$), which gives the second highest T_0 , then suggests that the physical mechanism causing the large values of macroturbulence likely originates in stellar layers with $100\,000 \text{ K} \lesssim T \lesssim 200\,000 \text{ K}$. This is consistent with a physical origin of O-star photospheric macroturbulence in the iron-peak opacity zone located roughly at $T \approx 160\,000 \text{ K}$.

5 DISCUSSION AND CONCLUSIONS

The central result of this paper is that strong macroturbulence is present in the photospheres of all known slowly rotating magnetic O-stars except for NGC 1624–2, which has (by far) the strongest surface field, and in which such non-thermal line-broadening seems to be largely suppressed. Moreover, the broad lines observed in the other magnetic O-stars are typically quite similar to those seen in non-magnetic stars, suggesting a common origin of macroturbulence. As in non-magnetic stars (e.g., Markova et al. 2011), it is not clear why these less extreme magnetic stars show such a large range of macroturbulent velocities.

Using these unique constraints derived from the magnetic O-stars, we assume (in analogy with Ap stars) that the organized, presumed fossil field can inhibit large-scale atmospheric motions down to stellar layers where the magnetic and gas pressures are comparable. The analysis here then indicates that the physical mechanism responsible for macroturbulence in O-stars likely originates in atmospheric layers with temperatures corresponding to $100\,000 \text{ K} \lesssim T \lesssim 200\,000 \text{ K}$.

An attractive scenario then is that this physical mechanism may be stellar oscillations excited by convection in the near-surface, iron-bump opacity zone of hot stars (Shiode et al. 2013). Future theoretical studies should calculate the excitation of such pulsational modes and their effects on spectral line formation, extending the work by Aerts et al. (2009) who computed line profiles from the collective effect of more deep-seated g-mode pulsations in evolved hot stars.

Finally, we also note that the iron opacity bump becomes weaker for lower metallicity (Cantiello et al. 2009). Thus, if indeed sub-surface convection is ultimately responsible for the observed non-thermal line broadening in early-type main-sequence stars, it would imply that lower metallicity stars should have lower macroturbulence. This should be observationally examined by comparing inferred macroturbulent velocities in the Galaxy and the Magellanic Clouds.

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